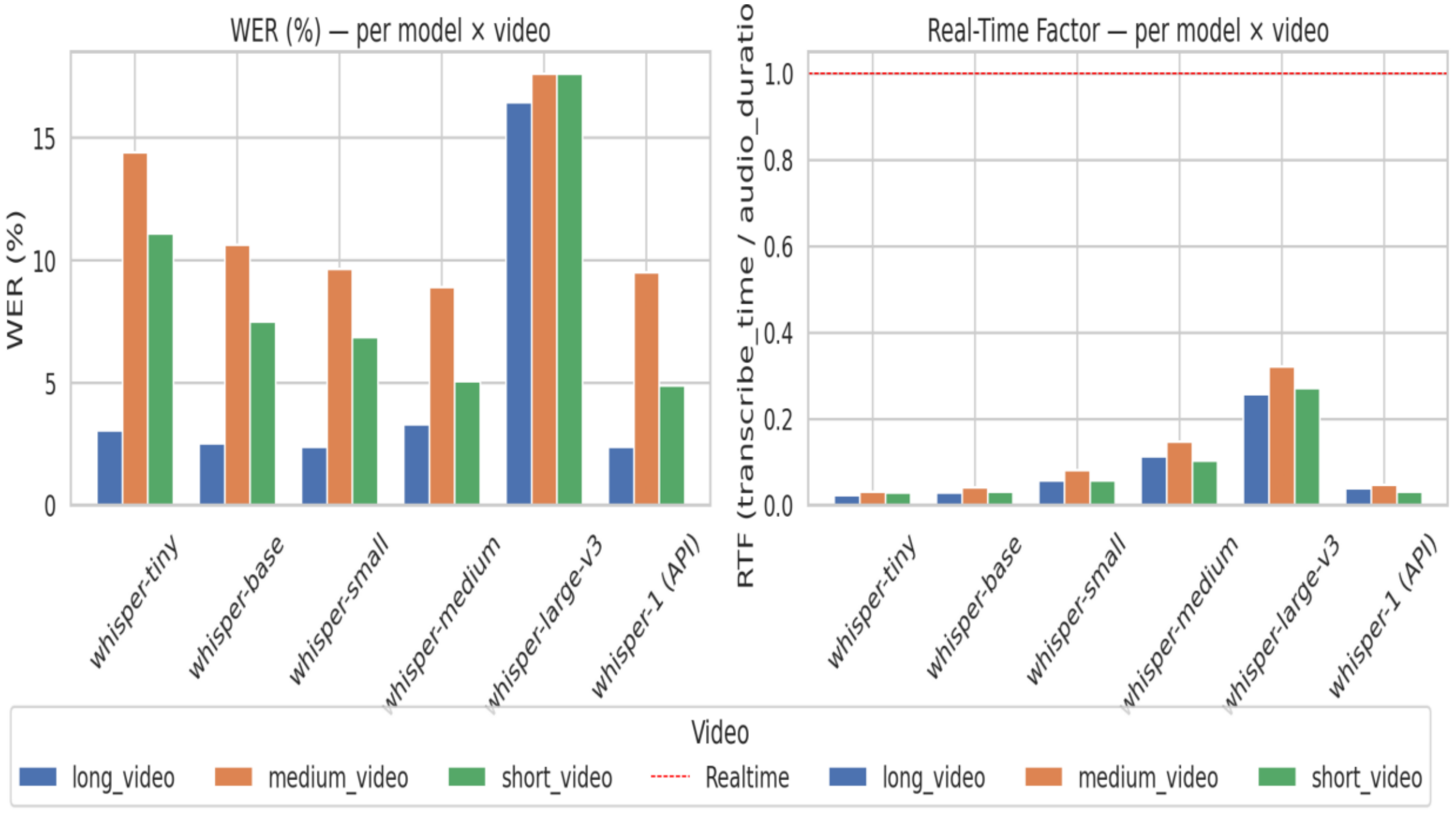
YouTube QA Bot — Benchmark Report

Ironhack AI Bootcamp — Final Project

Benchmarks: 7 | Videos: 3 | Whisper models: 6

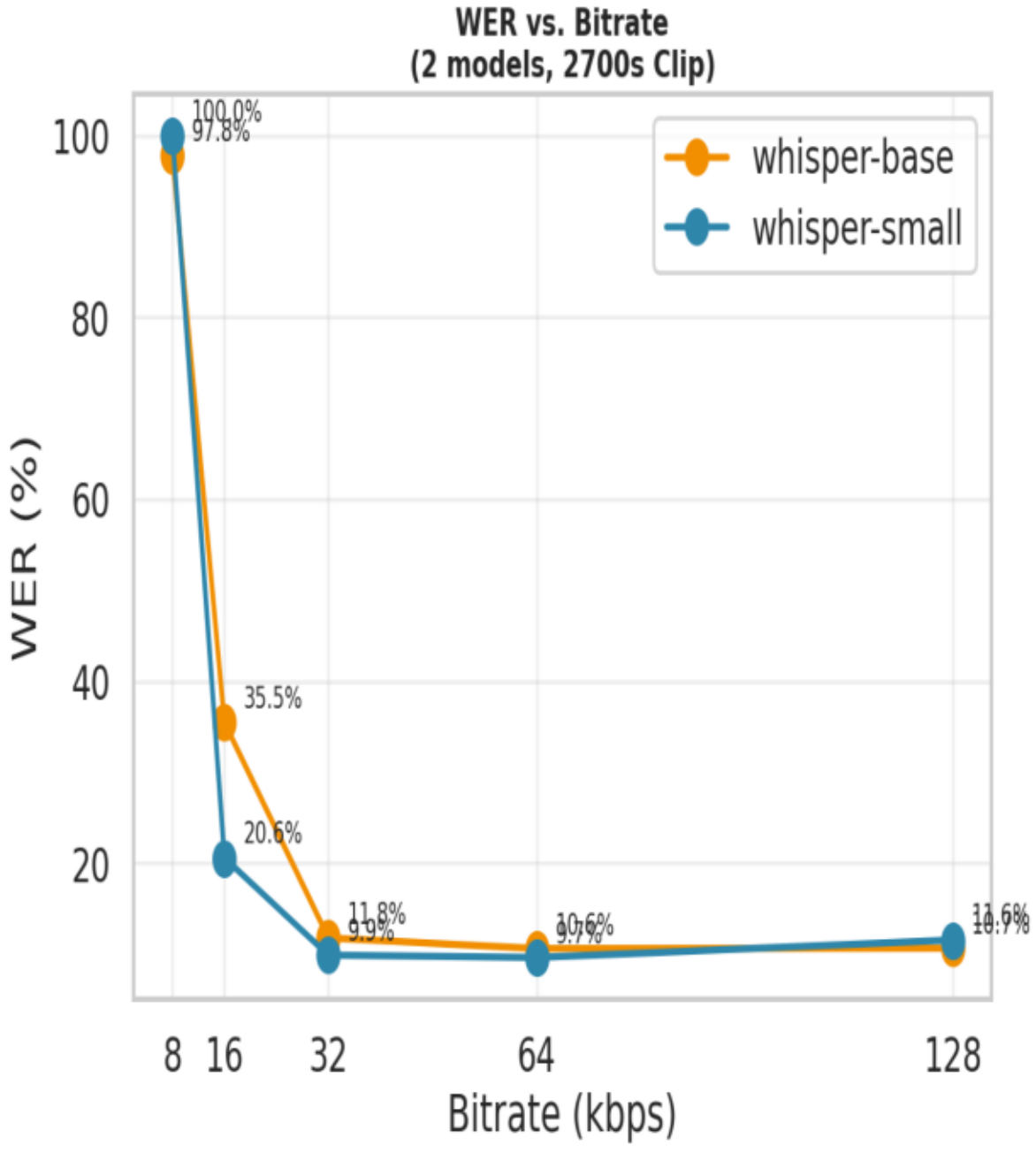
Benchmark 1: Whisper Model Sizes — Plot

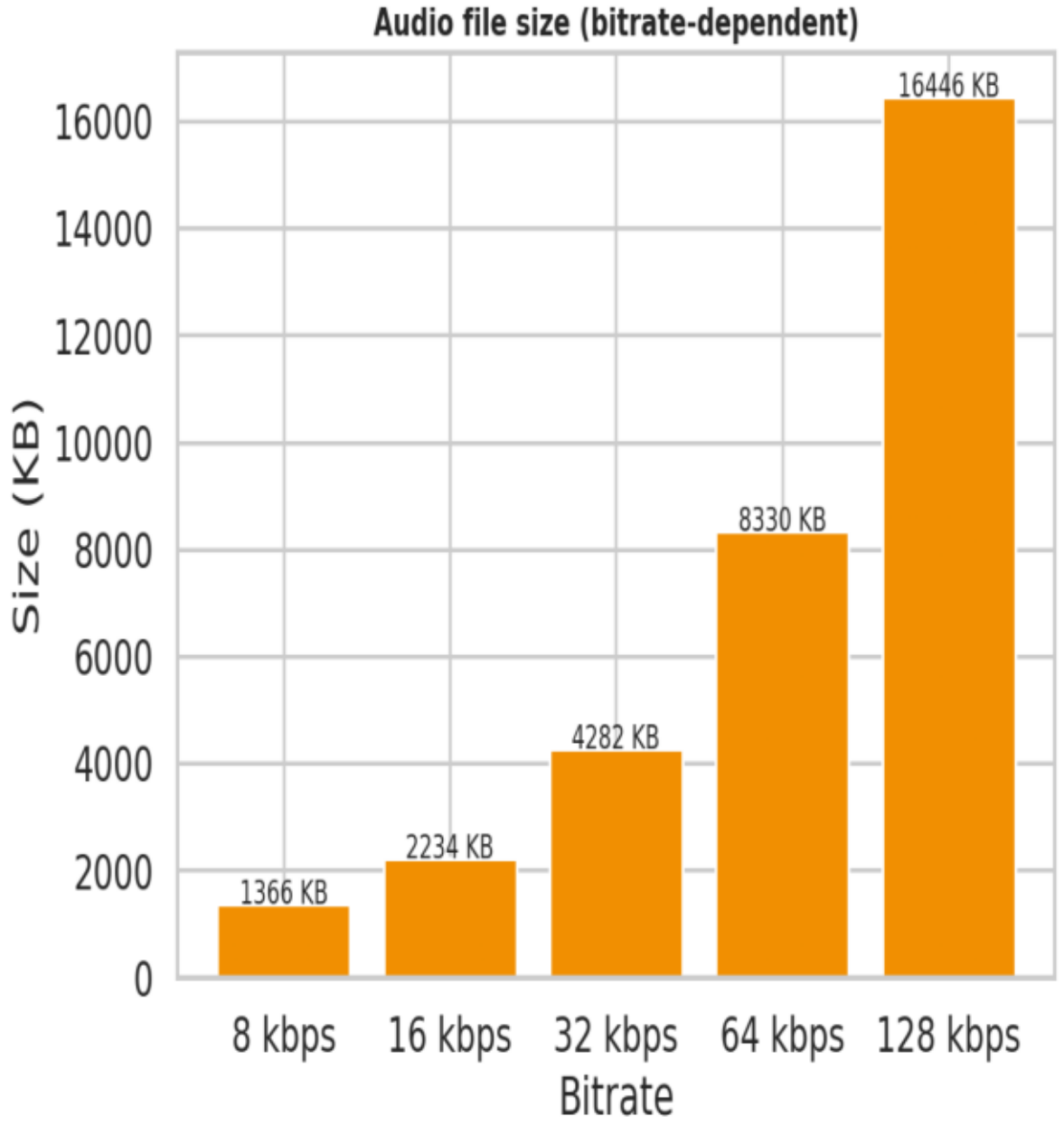


Benchmark 1: Whisper Model Sizes — Data

model	type	load_time_sec	transcribe_time_sec	total_time_sec	wer_percent	cost_usd	detected_language	video_label	video_id	video_duration_sec	color
whisper-tiny	open-source	2.22	9.57	11.78	11.09	0.0003	en	short_video	l-rBd0F6d9E	328	#2E86AB
whisper-base	open-source	4.1	10.5	14.61	7.5	0.0004	en	short_video	l-rBd0F6d9E	328	#2E86AB
whisper-small	open-source	14.35	19.08	33.42	6.85	0.0009	en	short_video	l-rBd0F6d9E	328	#2E86AB
whisper-medium	open-source	32.6	33.89	66.49	5.06	0.0018	en	short_video	l-rBd0F6d9E	328	#2E86AB
whisper-large-v3	open-source	127.17	89.12	216.29	17.62	0.006	en	short_video	l-rBd0F6d9E	328	#2E86AB
whisper-1 (API)	openai-api	0.0	10.5	10.5	4.89	0.0328	english	short_video	l-rBd0F6d9E	328	#F18F01
whisper-tiny	open-source	0.84	32.65	33.49	14.41	0.0009	en	medium_video	ubjW4aq_MAg	1037	#2E86AB
whisper-base	open-source	1.65	43.44	45.1	10.64	0.0012	en	medium_video	ubjW4aq_MAg	1037	#2E86AB
whisper-small	open-source	5.79	83.41	89.21	9.65	0.0025	en	medium_video	ubjW4aq_MAg	1037	#2E86AB
whisper-medium	open-source	21.98	152.57	174.55	8.92	0.0048	en	medium_video	ubjW4aq_MAg	1037	#2E86AB
whisper-large-v3	open-source	43.98	333.36	377.33	17.61	0.0105	en	medium_video	ubjW4aq_MAg	1037	#2E86AB
whisper-1 (API)	openai-api	0.0	48.73	48.73	9.52	0.1037	english	medium_video	ubjW4aq_MAg	1037	#F18F01
whisper-tiny	open-source	0.88	57.25	58.13	3.07	0.0016	en	long_video	B5K3Wj6r_HU	2538	#2E86AB
whisper-base	open-source	1.89	76.28	78.17	2.52	0.0022	en	long_video	B5K3Wj6r_HU	2538	#2E86AB
whisper-small	open-source	4.68	144.24	148.91	2.4	0.0041	en	long_video	B5K3Wj6r_HU	2538	#2E86AB
whisper-medium	open-source	21.65	287.4	309.05	3.31	0.0086	en	long_video	B5K3Wj6r_HU	2538	#2E86AB
whisper-large-v3	open-source	41.85	652.97	694.82	16.44	0.0193	en	long_video	B5K3Wj6r_HU	2538	#2E86AB
whisper-1 (API)	openai-api	0.0	99.88	99.88	2.38	0.2538	english	long_video	B5K3Wj6r_HU	2538	#F18F01

Benchmark 2: Audio Bitrate — Plot

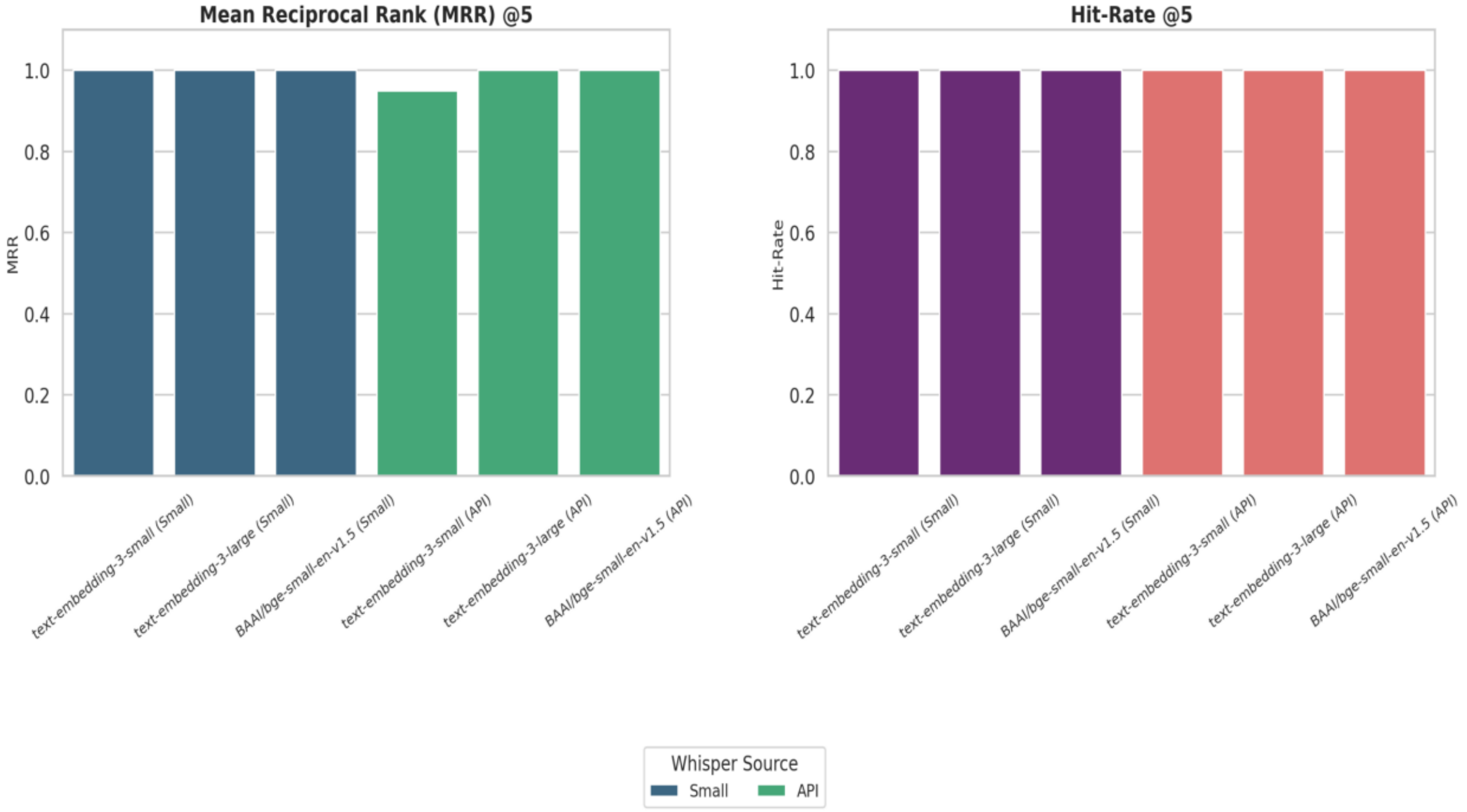




Benchmark 2: Audio Bitrate — Data

model	type	load_time_sec	transcribe_time_sec	total_time_sec	wer_percent	cost_usd	detected_language	bitrate_kbps	model_size	audio_size_kb
whisper-base [8 kbps	open-source	1.27	350.92	352.19	97.75	0.0098	la	8	base	1365.5
whisper-base [16 kbp	open-source	1.54	46.45	47.99	35.46	0.0013	en	16	base	2234.1
whisper-base [32 kbp	open-source	1.71	42.43	44.14	11.8	0.0012	en	32	base	4281.5
whisper-base [64 kbp	open-source	1.17	43.85	45.02	10.64	0.0012	en	64	base	8329.6
whisper-base [128 kb	open-source	1.14	43.48	44.61	10.67	0.0012	en	128	base	16446.3
whisper-small [8 kbp	open-source	4.32	763.24	767.56	99.97	0.0213	mn	8	small	1365.5
whisper-small [16 kb	open-source	3.68	82.78	86.46	20.59	0.0024	en	16	small	2234.1
whisper-small [32 kb	open-source	3.7	85.47	89.17	9.91	0.0025	en	32	small	4281.5
whisper-small [64 kb	open-source	3.76	83.66	87.42	9.65	0.0024	en	64	small	8329.6
whisper-small [128 k	open-source	3.63	83.95	87.58	11.57	0.0024	en	128	small	16446.3

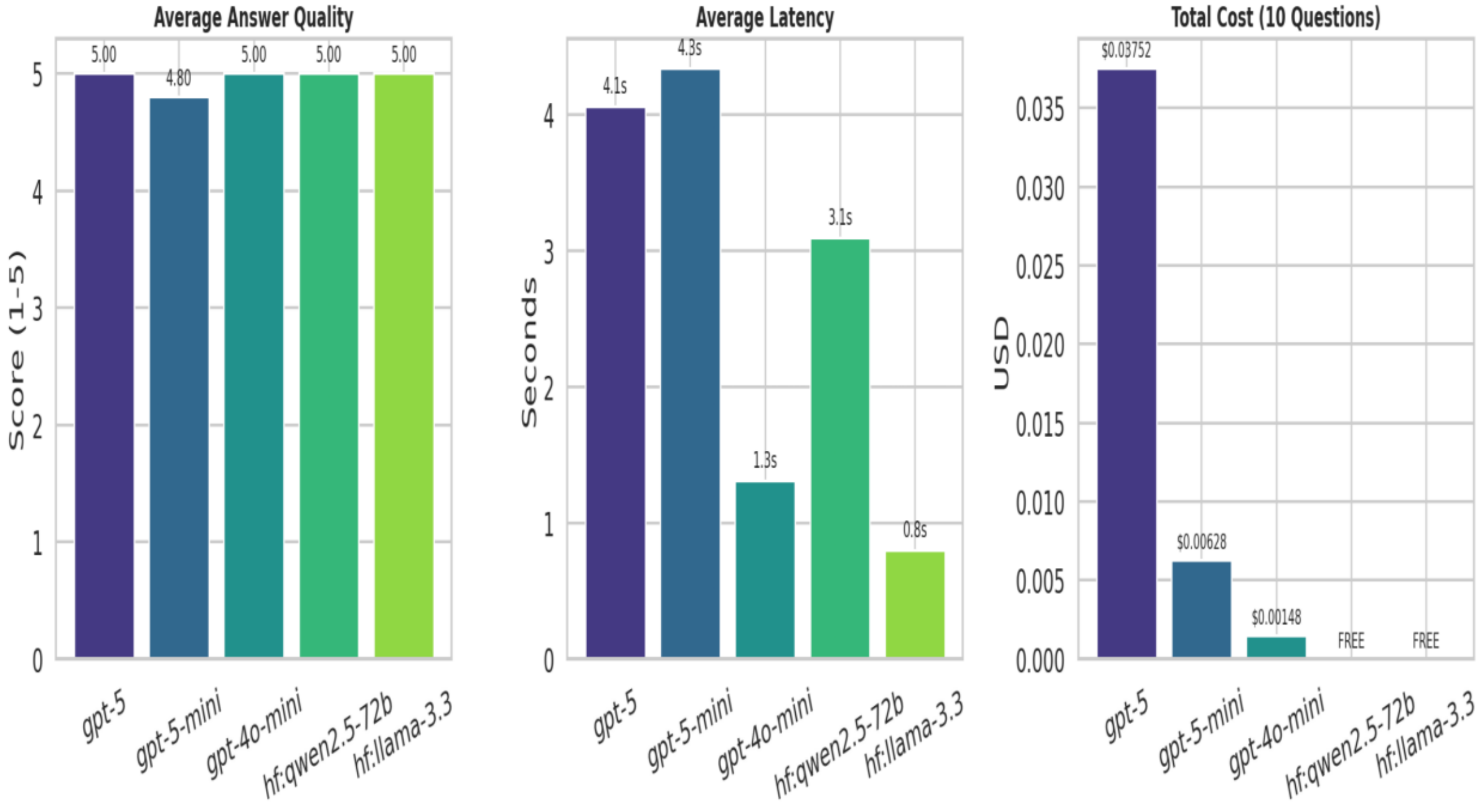
Benchmark 3: Embedding Models — Plot



Benchmark 3: Embedding Models — Data

model	hit_rate_at_k	mean_mrr	total_time_sec	embed_cost_usd	whisper_source
text-embedding-3-small (Small)	1.0	1.0	2.37	0.0001	Small
text-embedding-3-large (Small)	1.0	1.0	1.88	0.0006	Small
BAAI/bge-small-en-v1.5 (Small)	1.0	1.0	0.45	0.0	Small
text-embedding-3-small (API)	1.0	0.95	1.07	0.0001	API
text-embedding-3-large (API)	1.0	1.0	1.36	0.0006	API
BAAI/bge-small-en-v1.5 (API)	1.0	1.0	0.18	0.0	API

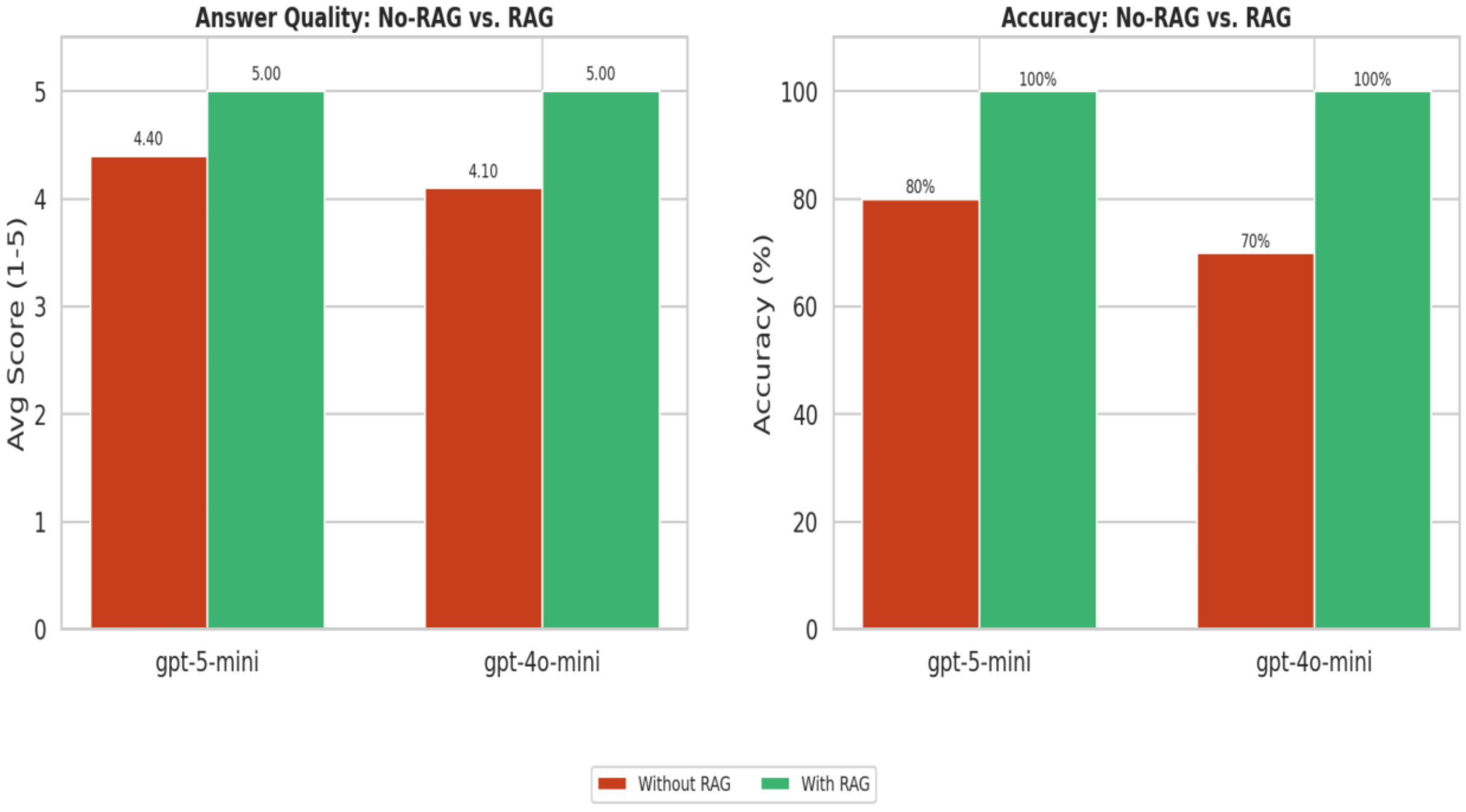
Benchmark 4: LLM Comparison — Plot



Benchmark 4: LLM Comparison — Data

model	avg_score	accuracy	avg_latency_sec	total_cost_usd	judge_fail_count
gpt-5	5.0	1.0	4.06	0.0375	0
gpt-5-mini	4.8	0.9	4.34	0.0063	0
gpt-4o-mini	5.0	1.0	1.31	0.0015	0
hf:qwen2.5-72b	5.0	1.0	3.09	0.0	0
hf:llama-3.3	5.0	1.0	0.8	0.0	0

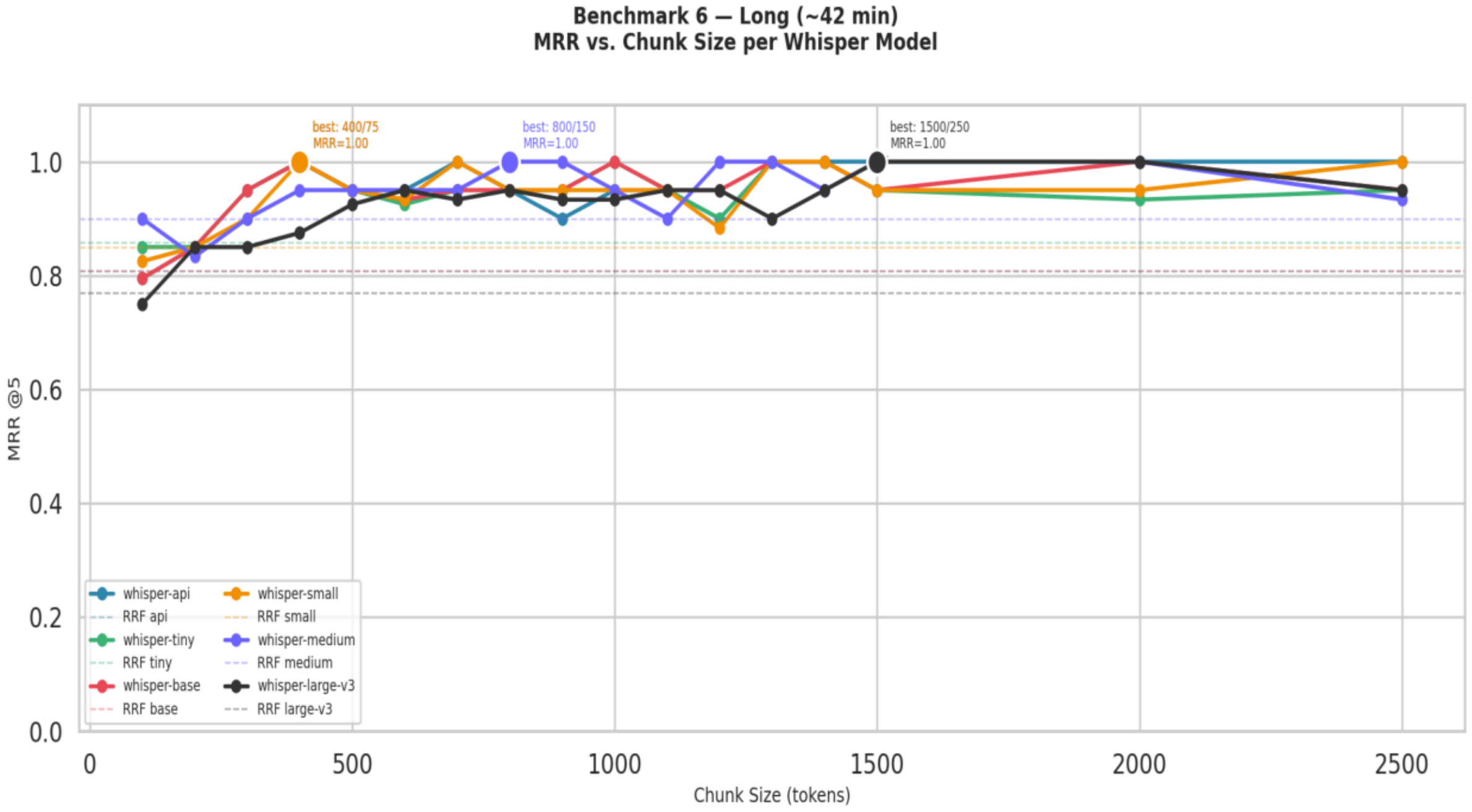
Benchmark 5: RAG vs. No-RAG — Plot



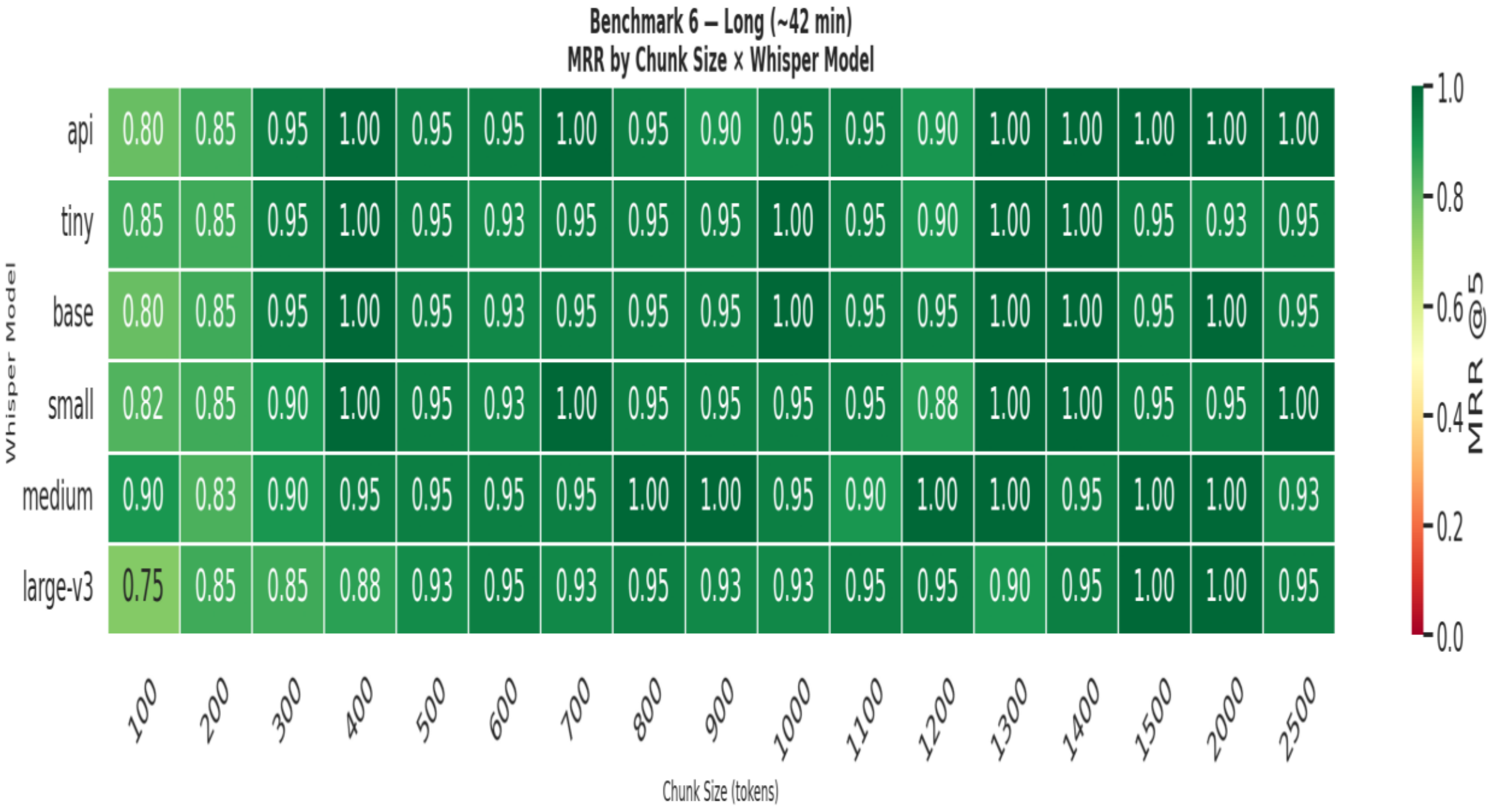
Benchmark 5: RAG vs. No-RAG — Data

model	variant	avg_score	accuracy	avg_latency_sec	total_cost_usd
gpt-5-mini	no-RAG	4.4	0.8	7.2466	0.0064
gpt-5-mini	with-RAG	5.0	1.0	4.9475	0.0067
gpt-4o-mini	no-RAG	4.1	0.7	1.3602	0.0003
gpt-4o-mini	with-RAG	5.0	1.0	1.2764	0.0015

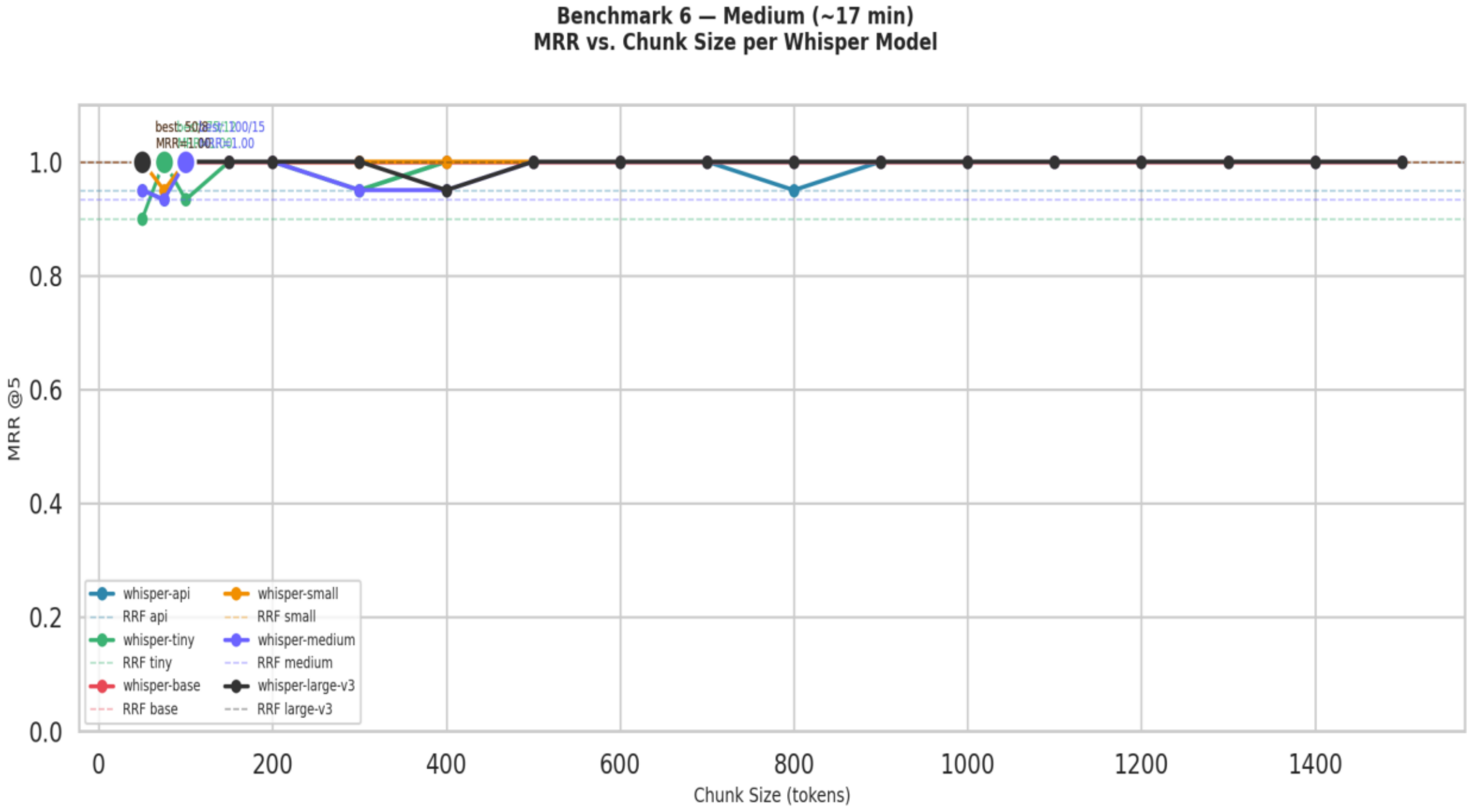
Benchmark 6: Chunk-Size Grid — benchmark6_long_video_curves



Benchmark 6: Chunk-Size Grid — benchmark6_long_video_heatmap

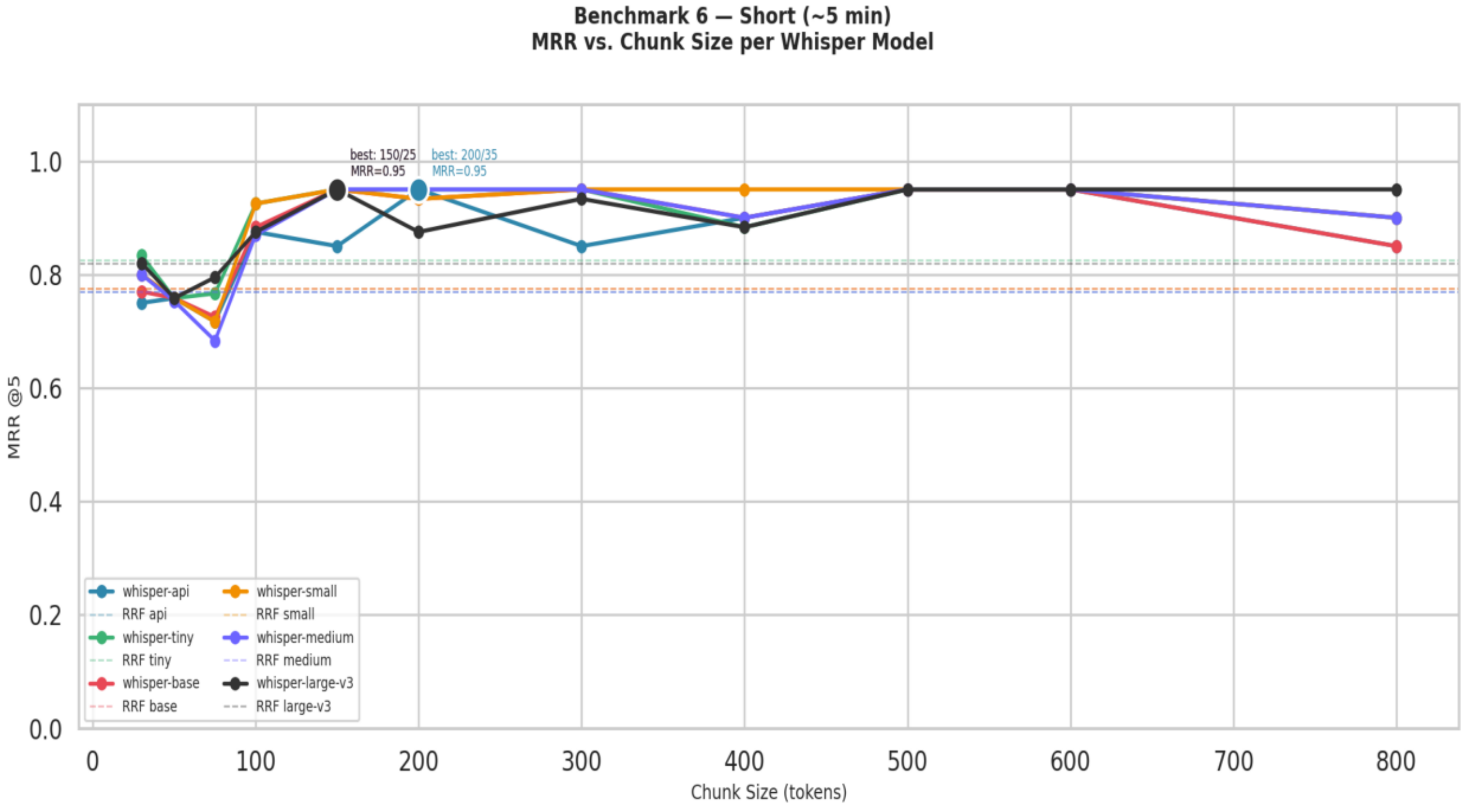


Benchmark 6: Chunk-Size Grid — benchmark6_medium_video_curves

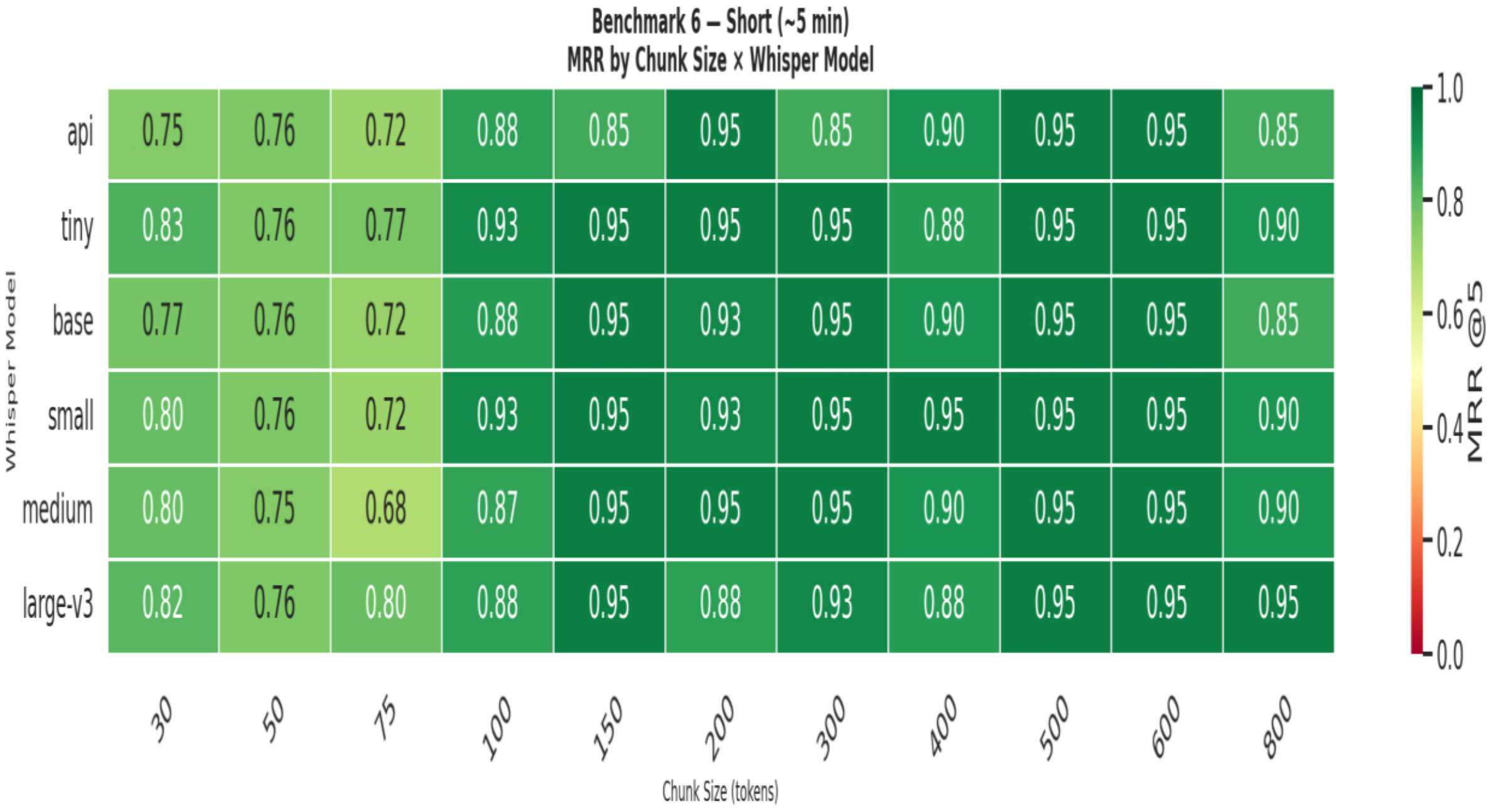


Benchmark 6: Chunk-Size Grid — benchmark6_medium_video_heatmap

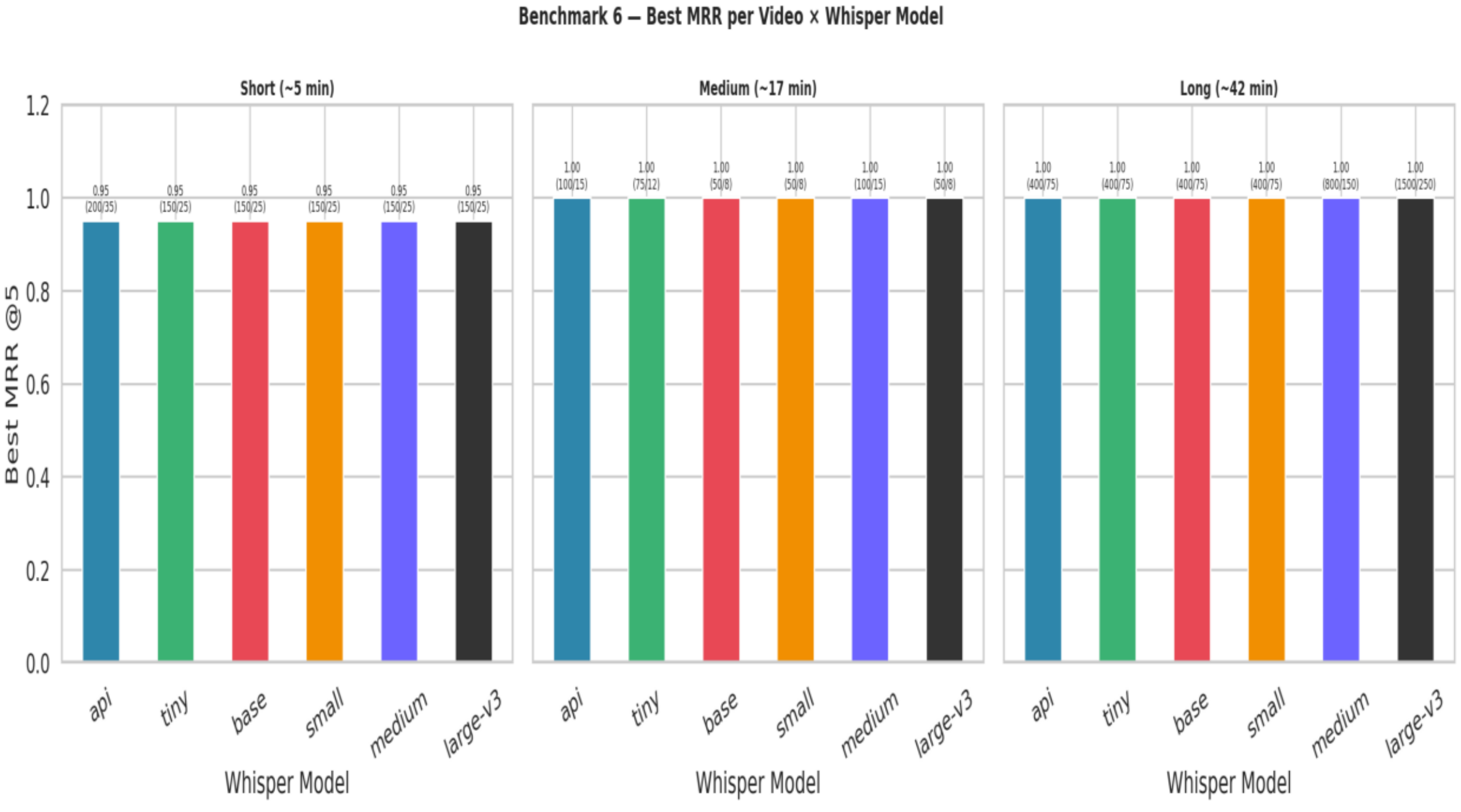
Benchmark 6: Chunk-Size Grid — benchmark6_short_video_curves



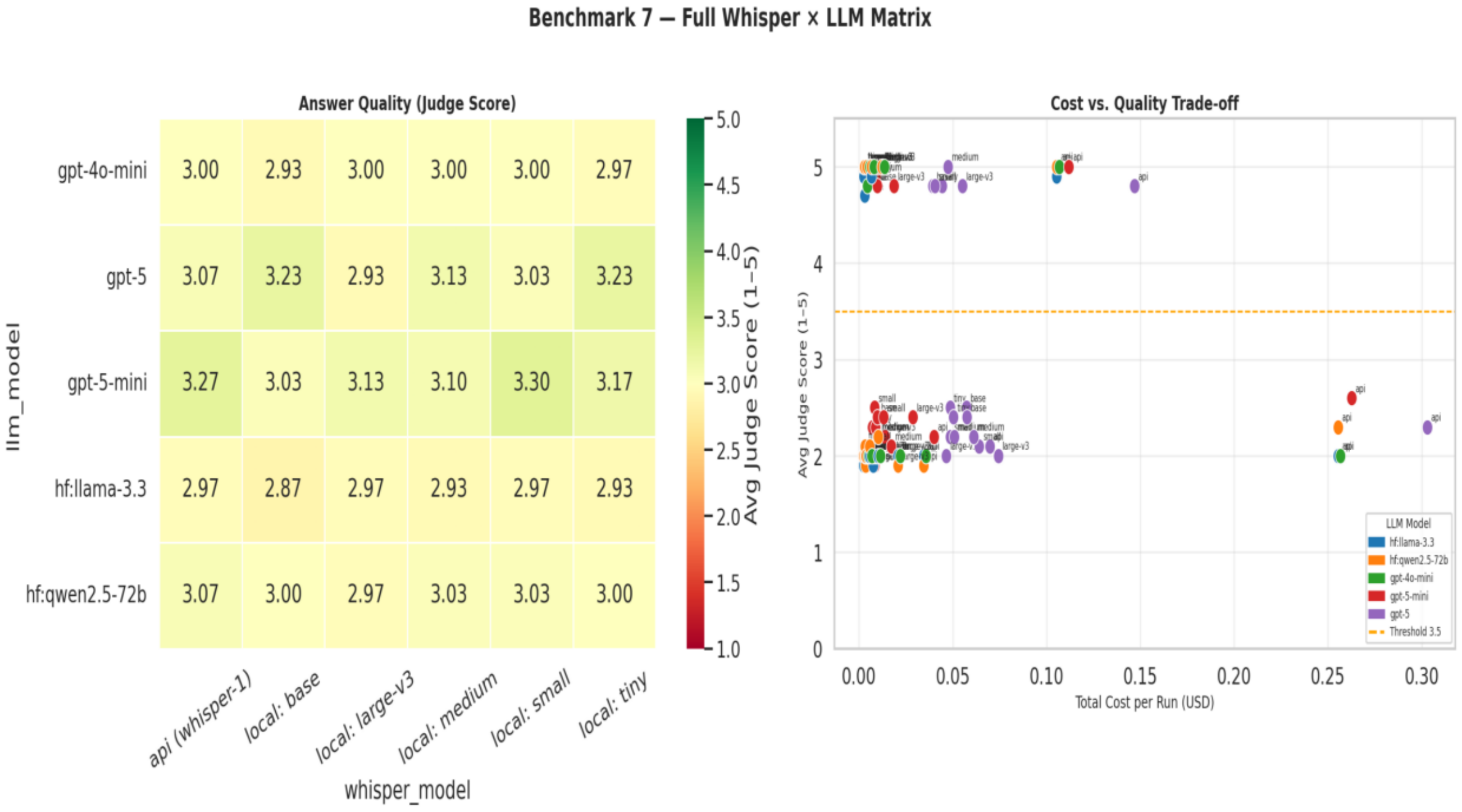
Benchmark 6: Chunk-Size Grid — benchmark6_short_video_heatmap



Benchmark 6: Chunk-Size Grid — benchmark6_summary



Benchmark 7: Full Whisper × LLM Matrix — Plot



Benchmark 7: Full Whisper × LLM Matrix — Data

whisper_model	whisper_wer_percent	llm_model	llm_avg_score	llm_avg_latency_sec	llm_total_cost_usd	judge_cost_usd	combo_total_cost_usd	judge_thumbs	judge_fail_count
local: tiny	11.09	hf:llama-3.3	1.9	0.53	0.0	0.0017	0.002	down	0
local: tiny	11.09	hf:qwen2.5-72b	1.9	1.24	0.0	0.0017	0.002	down	0
local: tiny	11.09	gpt-4o-mini	1.9	1.76	0.0013	0.0017	0.0033	down	0
local: tiny	11.09	gpt-5-mini	2.3	4.61	0.0051	0.0017	0.0071	down	0
local: tiny	11.09	gpt-5	2.5	5.65	0.0466	0.0017	0.0486	down	0
local: base	7.5	hf:llama-3.3	1.9	0.81	0.0	0.0017	0.0021	down	0
local: base	7.5	hf:qwen2.5-72b	2.0	1.19	0.0	0.0017	0.0021	down	0
local: base	7.5	gpt-4o-mini	2.0	0.71	0.0013	0.0017	0.0034	down	0
local: base	7.5	gpt-5-mini	1.9	4.85	0.0062	0.0017	0.0083	down	0
local: base	7.5	gpt-5	2.5	7.02	0.0552	0.0017	0.0573	down	0
local: small	6.85	hf:llama-3.3	1.9	0.6	0.0	0.0017	0.0026	down	0
local: small	6.85	hf:qwen2.5-72b	2.0	1.22	0.0	0.0017	0.0026	down	0
local: small	6.85	gpt-4o-mini	2.0	0.47	0.0012	0.0017	0.0038	down	0
local: small	6.85	gpt-5-mini	2.5	4.67	0.0057	0.0017	0.0083	down	0
local: small	6.85	gpt-5	2.2	5.12	0.0461	0.0017	0.0487	down	0
local: medium	5.06	hf:llama-3.3	1.9	0.51	0.0	0.0017	0.0035	down	0
local: medium	5.06	hf:qwen2.5-72b	1.9	1.23	0.0	0.0017	0.0035	down	0
local: medium	5.06	gpt-4o-mini	2.0	0.94	0.0013	0.0017	0.0048	down	0
local: medium	5.06	gpt-5-mini	2.2	5.62	0.0068	0.0017	0.0103	down	0
local: medium	5.06	gpt-5	2.2	5.28	0.0473	0.0017	0.0508	down	0
local: large-v3	17.62	hf:llama-3.3	1.9	0.44	0.0	0.0017	0.0077	down	0
local: large-v3	17.62	hf:qwen2.5-72b	2.0	1.47	0.0	0.0017	0.0077	down	0
local: large-v3	17.62	gpt-4o-mini	2.0	0.64	0.0014	0.0017	0.0091	down	0
local: large-v3	17.62	gpt-5-mini	2.2	7.49	0.0063	0.0017	0.014	down	0
local: large-v3	17.62	gpt-5	2.0	4.18	0.0389	0.0017	0.0465	down	0

Benchmark 6 — Best Config per Video × Whisper

video	whisper	chunk_size	overlap	mrr	hit_rate	rrf_mrr
short_video	api	200	35	0.95	1.0	0.77
medium_video	api	100	15	1.0	1.0	0.95
long_video	api	400	75	1.0	1.0	0.808
short_video	tiny	150	25	0.95	1.0	0.825
medium_video	tiny	75	12	1.0	1.0	0.9
long_video	tiny	400	75	1.0	1.0	0.858
short_video	base	150	25	0.95	1.0	0.775
medium_video	base	50	8	1.0	1.0	1.0
long_video	base	400	75	1.0	1.0	0.808
short_video	small	150	25	0.95	1.0	0.775
medium_video	small	50	8	1.0	1.0	1.0
long_video	small	400	75	1.0	1.0	0.85
short_video	medium	150	25	0.95	1.0	0.77
medium_video	medium	100	15	1.0	1.0	0.933
long_video	medium	800	150	1.0	1.0	0.9
short_video	large-v3	150	25	0.95	1.0	0.82
medium_video	large-v3	50	8	1.0	1.0	1.0
long_video	large-v3	1500	250	1.0	1.0	0.77

Benchmark 7 — Full Matrix Data

whisper_model	whisper_wer_percent	llm_model	llm_avg_score	llm_avg_latency_sec	llm_total_cost_usd	judge_cost_usd	combo_total_cost_usd	judge_thumbs	judge_fail_count
local: tiny	11.09	hf:llama-3.3	1.9	0.53	0.0	0.0017	0.002	down	0
local: tiny	11.09	hf:qwen2.5-72b	1.9	1.24	0.0	0.0017	0.002	down	0
local: tiny	11.09	gpt-4o-mini	1.9	1.76	0.0013	0.0017	0.0033	down	0
local: tiny	11.09	gpt-5-mini	2.3	4.61	0.0051	0.0017	0.0071	down	0
local: tiny	11.09	gpt-5	2.5	5.65	0.0466	0.0017	0.0486	down	0
local: base	7.5	hf:llama-3.3	1.9	0.81	0.0	0.0017	0.0021	down	0
local: base	7.5	hf:qwen2.5-72b	2.0	1.19	0.0	0.0017	0.0021	down	0
local: base	7.5	gpt-4o-mini	2.0	0.71	0.0013	0.0017	0.0034	down	0
local: base	7.5	gpt-5-mini	1.9	4.85	0.0062	0.0017	0.0083	down	0
local: base	7.5	gpt-5	2.5	7.02	0.0552	0.0017	0.0573	down	0
local: small	6.85	hf:llama-3.3	1.9	0.6	0.0	0.0017	0.0026	down	0
local: small	6.85	hf:qwen2.5-72b	2.0	1.22	0.0	0.0017	0.0026	down	0
local: small	6.85	gpt-4o-mini	2.0	0.47	0.0012	0.0017	0.0038	down	0
local: small	6.85	gpt-5-mini	2.5	4.67	0.0057	0.0017	0.0083	down	0
local: small	6.85	gpt-5	2.2	5.12	0.0461	0.0017	0.0487	down	0
local: medium	5.06	hf:llama-3.3	1.9	0.51	0.0	0.0017	0.0035	down	0
local: medium	5.06	hf:qwen2.5-72b	1.9	1.23	0.0	0.0017	0.0035	down	0
local: medium	5.06	gpt-4o-mini	2.0	0.94	0.0013	0.0017	0.0048	down	0
local: medium	5.06	gpt-5-mini	2.2	5.62	0.0068	0.0017	0.0103	down	0
local: medium	5.06	gpt-5	2.2	5.28	0.0473	0.0017	0.0508	down	0
local: large-v3	17.62	hf:llama-3.3	1.9	0.44	0.0	0.0017	0.0077	down	0
local: large-v3	17.62	hf:qwen2.5-72b	2.0	1.47	0.0	0.0017	0.0077	down	0
local: large-v3	17.62	gpt-4o-mini	2.0	0.64	0.0014	0.0017	0.0091	down	0
local: large-v3	17.62	gpt-5-mini	2.2	7.49	0.0063	0.0017	0.014	down	0
local: large-v3	17.62	gpt-5	2.0	4.18	0.0389	0.0017	0.0465	down	0